CS301: Data Structures and Algorithms - Syllabus 2025

Document: CSE-SYLLABUS-CS301 | Academic Year: 2025-2026 | Credits: 4 (3L + 1T + 2P)

Course Overview and Objectives:

CS301 delivers foundational knowledge of core computational structures, design techniques, and algorithm complexity. Prerequisites: CS101 Programming in C / C++ and MA102 Discrete Mathematics.

Module 1: Foundations and Asymptotic Analysis (8 Hours)

- Growth of Functions and Asymptotic notations: Big-O, Big-Omega, Big-Theta, Little-o, Little-omega.
- Recurrence Relations: Substitution method, Recursion Tree method, and Master Theorem.
- Abstract Data Types (ADTs): Stacks, Queues, Circular Queues, Deque, Priority Queues.
- Applications: Infix to Postfix expression conversion, evaluation, balanced parenthesis verification.

Module 2: Non-Linear Structures - Trees and Heaps (10 Hours)

- Binary Tree traversals (Inorder, Preorder, Postorder, Level order).
- Binary Search Trees (BST): Insertion, Deletion, Searching, Successor/Predecessor operations.
- Balanced BSTs: AVL Trees (Single and Double rotations), Red-Black Tree invariant properties.
- Multiway Search Trees: B-Trees and B+ Trees (node splitting, merging, indexing fundamentals).
- Priority Queues: Binary Min-Heap and Max-Heap, Heapify algorithm, HeapSort (O(n log n) runtime).

Module 3: Graph Representations and Traversal Algorithms (10 Hours)

- Representations: Adjacency Matrix, Adjacency List, Compressed Sparse Row (CSR).
- Graph Traversal: Breadth-First Search (BFS) and Depth-First Search (DFS), Classification of edges.
- Topological Sorting (Kahn's algorithm and DFS-based) and Cycle detection in directed/undirected graphs.
- Minimum Spanning Trees (MST): Kruskal's algorithm (Disjoint Set Union) and Prim's algorithm.
- Single-Source Shortest Paths: Dijkstra's Algorithm (with priority queue), Bellman-Ford (negative weight detection).
- All-Pairs Shortest Path: Floyd-Warshall Algorithm (O(V^3)).

Module 4: Algorithm Design Paradigms (10 Hours)

- Greedy Paradigm: Optimal substructure and greedy-choice property. Fractional Knapsack, Huffman coding.
- Dynamic Programming: Principle of Optimality, Memoization vs Tabulation. 0/1 Knapsack, Longest Common Subsequence (LCS), Matrix Chain Multiplication, Subset Sum problem.
- Divide and Conquer: MergeSort, QuickSort (randomized pivot selection, worst vs average case analysis).

Textbooks and Prescribed References:

1. Primary Textbook: 'Introduction to Algorithms' by Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, and Clifford Stein (CLRS), 4th Edition, MIT Press, 2022.
2. Reference Book 1: 'Data Structures and Algorithm Analysis in C++' by Mark Allen Weiss, 4th Edition, Pearson.
3. Reference Book 2: 'Algorithm Design' by Jon Kleinberg and Eva Tardos, 1st Edition, Pearson.

Assessment and Examination Scheme:

- Continuous Internal Assessment (CIA): 30% (Mid-Semester Exam)
- Laboratory Practical Assignments & Code Reviews: 20%
- Course Capstone Project: 10%
- End-Semester Final Examination: 40% (Comprehensive Theory, 3 Hours Duration)
- Passing Minimum: 40% overall aggregate marks across theory and laboratory components.